

training reasoning would be wrong. Training on incorrect reasoning would teach wrong decision boundaries.

By providing the correct answer, we decouple the teacher’s classification ability from its description ability. GPT-4.1 can describe visual features competently even on tasks where it classifies poorly. This decoupling is what allows the student to outperform the teacher: the student inherits correct reasoning patterns without inheriting the teacher’s classification errors.

6.3 The Steel Surface Gap

On steel surface defects, GPT-4.1 few-shot (91.1%) outperforms our CoT-augmented model (66.7%). This is the only task where the frontier model maintains a large advantage. The likely reason is that GPT-4.1’s few-shot setting provides reference images for each class, enabling direct visual comparison. For 200×200 grayscale images where defect classes differ primarily in texture patterns, this visual reference is highly effective.

Our method does not provide reference images at inference time. The model must rely on learned internal representations. With only 5 training images per class (30 total), there may be insufficient coverage of the visual variation within each class. This suggests that the method’s effectiveness depends partly on the ratio of training examples to intra-class visual diversity.

6.4 Limitations

Single base model. All experiments use Qwen2.5-VL-3B [14]. We do not evaluate whether the method generalizes to other VLM architectures or model sizes. A larger model (7B or above) might benefit less from CoT augmentation if it already has sufficient capacity for direct learning.

Single seed per task. Due to compute constraints, each configuration was trained with a single random seed (42). We cannot report confidence intervals or statistical significance. The +0.9pp improvement on UHCS, in particular, may not be reliable without multiple runs.

No ablation on augmentation ratio. We use 3 CoT descriptions per image in all experiments. The optimal number may vary by task. More descriptions could help on harder tasks, or could introduce diminishing returns on easier ones.

Fixed training epochs. All models train for 40 epochs regardless of dataset size. This means the direct models (18–30 examples) see each image ~ 40 times, while the augmented models (72–120 examples) see each image ~ 10 times (across its 4 variants). Optimizing epochs per approach could improve both baselines.

7 Conclusion

We introduced answer-conditioned chain-of-thought distillation for rapidly adapting small vision-language models to industrial inspection tasks. The method uses a frontier VLM to generate justified visual reasoning for each training image, conditioned on the correct label, then fine-tunes a 3B-parameter model on these reasoning-augmented examples via LoRA.

Across four industrial classification tasks spanning three image modalities, the method consistently outperforms direct fine-tuning (+0.9 to +8.3 percentage points). On weld radio-graph classification, the fine-tuned 3B model outperforms GPT-4.1 few-shot by 10.8 percent-